

# **LAPORAN PRAKTIKUM TEKNOLOGI IOT II**

## **3.5.1 Praktikum**

*Dosen Pengampu:*

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**PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET**

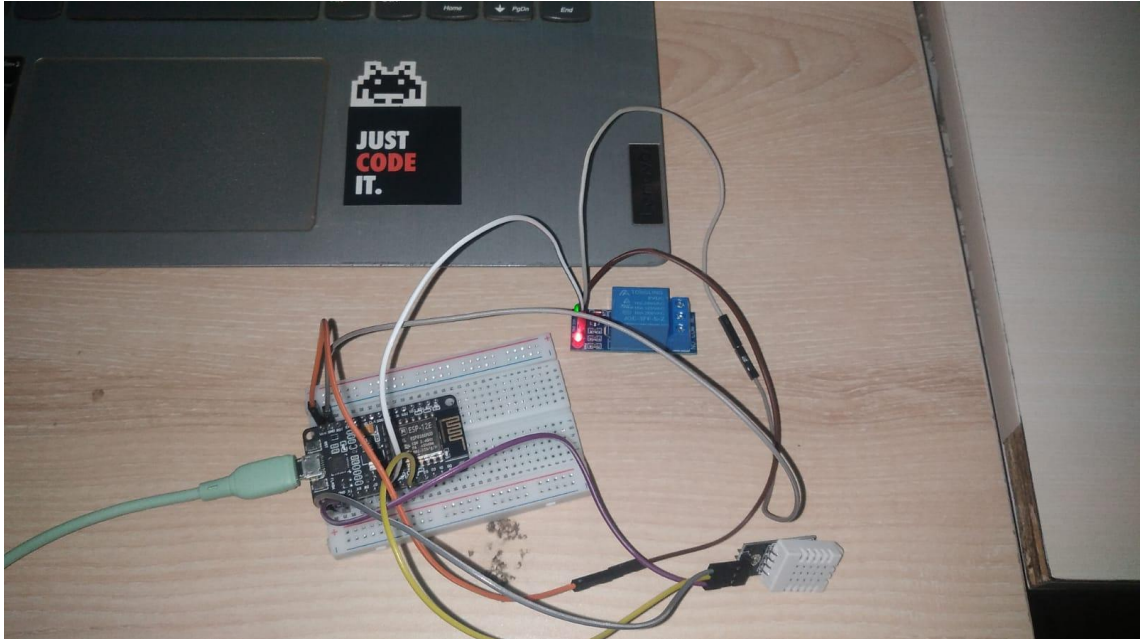
**JURUSAN TEKNOLOGI INFORMASI**

**POLITEKNIK NEGERI LAMPUNG**

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### 3.5.1 Praktikum 1: NodeMCU sebagai Web Server (Menyajikan HTML)

Praktik:



Web server :



#### ESP8266 Web Server

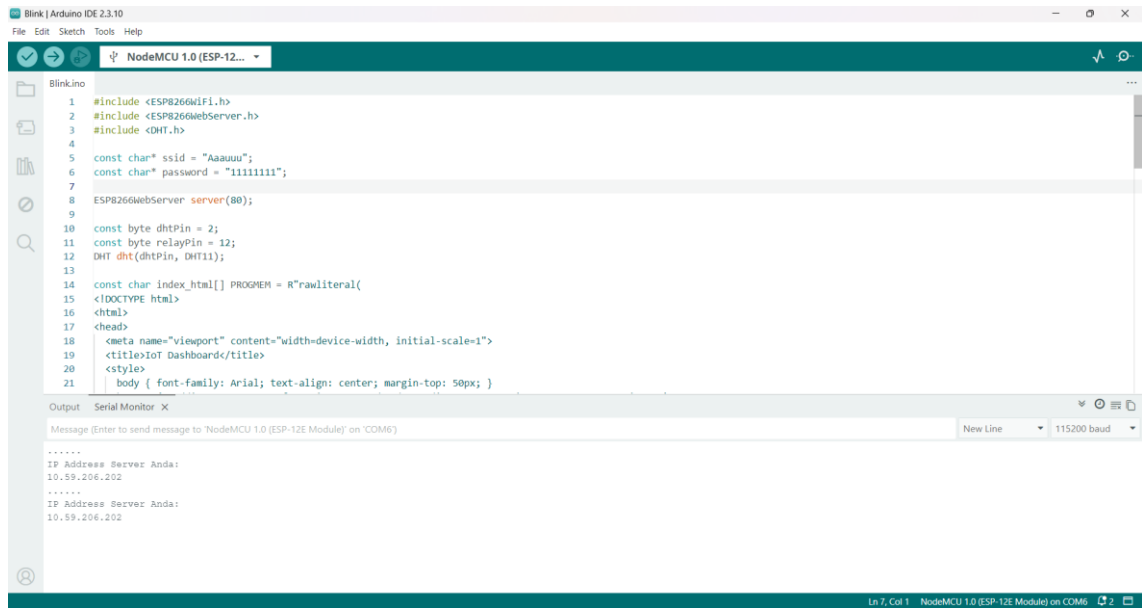
Suhu Saat Ini: **1.00** Celcius

Kendali Relay

ON

OFF

## Ardiuno IDE:



```
Blinkino
1 #include <ESP8266WiFi.h>
2 #include <ESP8266WebServer.h>
3 #include <DHT.h>
4
5 const char* ssid = "Aaauuu";
6 const char* password = "11111111";
7
8 ESP8266WebServer server(80);
9
10 const byte dhtPin = 2;
11 const byte relayPin = 12;
12 DHT dht(dhtPin, DHT11);
13
14 const char index_html[] PROGMEM = R"rawliteral(
15 <!DOCTYPE html>
16 <html>
17 <head>
18   <meta name="viewport" content="width=device-width, initial-scale=1">
19   <title>IoT Dashboard</title>
20   <style>
21     body { font-family: Arial; text-align: center; margin-top: 50px; }
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM6')

New Line 115200 baud

```
.....
IP Address Server Anda:
10.59.206.202
.....
IP Address Server Anda:
10.59.206.202
```

Ln 7, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM6 2

## Code:

```
#include <ESP8266WiFi.h>
```

```
#include <ESP8266WebServer.h>
```

```
#include <DHT.h>
```

```
const char* ssid = "Aaauuu";
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const char* password = "11111111";
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```
ESP8266WebServer server(80);
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```
const byte dhtPin = 2;
```

```
const byte relayPin = 12;
```

```
DHT dht(dhtPin, DHT11);
```

```
const char index_html[] PROGMEM = R"rawliteral(
```

```
<!DOCTYPE html>
```

```

<html>
<head>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>IoT Dashboard</title>
  <style>
    body { font-family: Arial; text-align: center; margin-top: 50px; }
    button { padding: 15px 30px; font-size: 20px; border-radius: 8px; margin: 10px;
cursor: pointer;}
    .btn-on { background-color: #4CAF50; color: white; border: none; }
    .btn-off { background-color: #f44336; color: white; border: none; }
    .sensor-box { font-size: 24px; font-weight: bold; }
  </style>
</head>
<body>
  <h1>ESP8266 Web Server</h1>
  <div class="sensor-box">
    <p>Suhu Saat Ini: <strong>%TEMPERATURE%</strong> Celcius</p>
  </div>
  <h2>Kendali Relay</h2>
  <a href="/relay/on"><button class="btn-on">ON</button></a>
  <a href="/relay/off"><button class="btn-off">OFF</button></a>
</body>
</html>
)rawliteral";

```

```

void handleRoot() {
  String html = index_html; // Salin kerangka HTML ke variabel dinamis
  float t = dht.readTemperature();

  // Mengganti teks placeholder %TEMPERATURE% dengan suhu nyata

```

```
if (isnan(t)) {  
    html.replace("%TEMPERATURE%", "--");  
} else {  
    html.replace("%TEMPERATURE%", String(t));  
}
```

```
server.send(200, "text/html", html);  
}
```

```
void handleRelayOn() {  
    digitalWrite(relayPin, HIGH);  
    server.sendHeader("Location", "/");  
    server.send(303);  
}
```

```
void handleRelayOff() {  
    digitalWrite(relayPin, LOW);  
    server.sendHeader("Location", "/");  
    server.send(303);  
}
```

```
void setup() {  
    Serial.begin(115200);  
    pinMode(relayPin, OUTPUT);  
    digitalWrite(relayPin, LOW);  
    dht.begin();
```

```
    WiFi.mode(WIFI_STA);  
    WiFi.begin(ssid, password);
```

```

while (WiFi.status() != WL_CONNECTED) { delay(500); Serial.print("."); }

Serial.println("\nIP Address Server Anda: ");

Serial.println(WiFi.localIP());


server.on("/", handleRoot);

server.on("/relay/on", handleRelayOn);

server.on("/relay/off", handleRelayOff);

server.begin();

}


void loop() {
    server.handleClient();
}

```

#### Penjelasan Singkat Kode:

- `ESP8266WebServer server(80);`: Menginisialisasi *library Server* agar terus mendengarkan panggilan di jalur lalu lintas standar web (*Port 80*).
- `html.replace("%TEMPERATURE%", String(t));`: Fitur manipulasi teks (*String*). Perintah ini secara cerdas akan menyapu dokumen HTML, mencari teks *placeholder* `%TEMPERATURE%`, dan menggantinya dengan variabel angka suhu yang didapat dari sensor aktual sebelum dikirim ke layar pengunjung. Inilah esensi dari halaman *web* dinamis.
- `server.on("/relay/on", handleRelayOn);`: Fungsi *Routing*. Jika ada klien mengunjungi tautan URL berakhiran `/relay/on`, server akan menjalankan aksi mengubah *state* Relay, lalu me- *redirect* layar pengunjung kembali ke halaman depan (Location: `/`) via kode 303.
- `server.handleClient();`: Berada di dalam `loop()` untuk terus-menerus mengecek apakah ada permintaan pengunjung yang datang setiap milidetiknya.